Date: Total Pages-3 Roll No.....

END SEMESTER EXAMINATION, MAY 2010
B.E. COE VI SEMESTER
COE 315 ADVANCED COMPUTER ARCHITECTURE

Duration : 3hours **Maximum Marks: 70**

Instructions:

1. Attempt any five questions
2. All questions carry equal marks (14). Marks of sub-parts are given.

Q1) a) A software application has been subjected to extensive simulation and the various branch probabilities have been evaluated. Which of the following instruction scheduling techniques may be suitable for implementing this application? Justify your answer.
(i) Dynamic pipeline scheduling using tagged reservation stations, tagged operands and register renaming (ii) Static trace scheduling on VLIW architecture.

(7)

Q1) b) Discuss bus dynamic network and multi-stage dynamic network in terms of switching complexity, delay, reliability and bandwidth per node. (7)

Q2) a) A group of modules comprising an application, interact very closely with each other at a fine granularity level. Which of the following inter processor communication models should be adopted when the tasks are implemented on a multiprocessor MIMD system? Give reasons. (i) shared memory on static network (ii) message passing on dynamic network.

(7)

Q2)b) Explain, with examples the main characteristics of dataflow architectures and their advantages w.r.t control flow architectures.

(7)

Q3) a) Explain how prefetch buffers, target buffers and loop buffers boost pipeline performance.

(5)

Q3) b) A digital system has been designed to perform 5 tasks in sequence. The times taken for the five tasks are as given:
Task 1: 5 ns,
Task 2: 7 ns
Task 3: 10 ns
Task 4: 8 ns
Task 5: 3 ns
Calculate the speedup, efficiency and throughput when the same system is pipelined. Assume latch delay to be 2ns. Due to technological enhancement, the same pipeline is next superpipelined to a degree 4. Calculate the speedup relative to the base machine.

(9)

Q4)a) Calculate the speedup for evaluating the cumulative sum of the elements of an array of N numbers on a linear array of *p* processors.

(5)

Q4)b) Derive the dependence equation, show the applicable dependency test and optimize the following code fragment:

```
Do i = 1,N
  Do j = 3,N
        B[i,j] = B[i, j-2] + B[i,j-1] + C;
  End do
End do
Do k = 1,N
   A[k] = A[N-k+1]
Enddo
Do k = 1,N
   A[k] = 2*A[k] + R^k + C*k
Enddo
```

Note: R and C are constants.

(9)

Q5)a) A shared block of data in a 3-processor system follows the Write Once or MESI protocol. A write/read/replace event by processor P_i is denoted as $W_i/R_i/Z_i$. For the given sequence of events, complete the table given below by filling in the states of the three cache blocks and the memory copy for each event.

Use the following legends for filling states:

1. Valid : V
2. Reserved: R
3. Dirty: D
4. Invalid: I

As an example, the first entry has been filled in for your reference.

Event	Processor 1	Processor 2	Processor 3	Memory copy
R_2	V	I	I	V
W_1				
R_2				
R_3				
Z_2				
W_2				
W_2				
W_1				
R_3				
W_3				
Z_3				
R_2				
W_3				

(9)

Q5) b) Write the characteristics of the DSB or the weak memory consistency model. (5)

Q6) a) A memory intensive image processing application that can render telescopic images of outer space with varying resolutions has to be implemented on a parallel processing machine. Which speedup performance law is suitable for characterizing its multiprocessor implementation? Analyze the impact of the chosen model on workload distribution, total time taken and variation of speedup with degree of sequentiality. (7)

Q6)b) A program comprises the following modes of operation:
Sequential mode: 20%
Parallel mode with Degree of parallelism (DOP) equal to 2: 30%
Parallel mode with DOP equal to 4: 10%
Parallel mode with DOP equal to 9: 40%
The total Using Amdahl's Law, calculate the speedup when the above program is executed on a four processor system. (7)

Q7)a) Explain with illustration, any one of the following code optimization techniques:
(i) wavefronting (ii) software pipelining (iii) Local common sub-expression elimination (7)

Q7) b) Write notes on any one of the following topics: (7)

1. Applications of supercomputing
2. Grid / Cluster / Cloud Computing
3. Supercomputing : The Indian scenario

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